

Signed Local Spectral Envelopes for Search-Free ADMM Tuning in Diffeomorphic Image Registration

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Abstract

We derive and test signed, registration-specific spectral enclosures for ADMM and over-relaxed ADMM (oADMM). The exact reduced operator is $(1 - \alpha/2)I + (\alpha/2)C_H C_G$. For constant image gradients and a scalar transform-diagonalizable Sobolev regularizer, its modal factors reduce to curvature corners; the globally optimal joint (ρ, α) lies in a finite algebraic candidate set. For spatially varying rank-one Hessian blocks, we retain the exact regularizer and derive block disks, a commutator rectangle, and a sharper comparison-matrix rectangle based on three scalar $n \times n$ eigenproblems rather than the full $dn \times dn$ iteration. We then separate prediction from certification: exact pixelwise curvatures choose parameters algebraically, after which one comparison-matrix evaluation certifies the chosen iteration. In ten separated synthetic regimes this predict-then-certify method produced a rigorous bound below one in every case, never collapsed, and retained median spectral-radius gaps to full optimization of 0.057 overall and 0.027 over nine smooth cases. The certificates are conservative (median relative bound gap 1.29), but parameter quality and certified convergence hold simultaneously. Constant-gradient results are exact to machine precision; 18 broader stress cases verify every enclosure without violation. These results provide a positive search-free methodology while identifying certificate tightness, rather than parameter quality, as the remaining obstacle before real-data claims.

1 Introduction

Diffeomorphic image registration estimates an invertible, topology-preserving spatial transformation that aligns a moving image with a fixed image. Classical methods remain attractive because they provide explicit control over the deformation model, regularization, and numerical stopping criteria, and because they can be applied without training data. Representative approaches include DARTEL, diffeomorphic demons, large-deformation optimal-control formulations, and modern GPU-accelerated variational solvers [1, 13, 8, 12]. Their main practical limitation is computational cost.

Many diffeomorphic schemes update a velocity field through a sequence of convex, linearized optical-flow problems. A typical quadratic subproblem is

$$\min_{v=(v_x, v_y)} \frac{\mu}{2} \|I_x \odot v_x + I_y \odot v_y + I_t\|_2^2 + \frac{1}{2} \|Lv\|_2^2, \quad (1)$$

where I_x, I_y are spatial image derivatives, I_t is the current temporal or residual derivative, and L is a differential regularization operator. Splitting the data and regularization terms produces inexpensive pixelwise and Fourier-domain subproblems, making ADMM particularly suitable for GPU implementation [12, 10].

The penalty ρ and relaxation α strongly affect convergence. Fixed defaults, residual balancing, and adaptive spectral rules can be effective, but they do not exploit the unusually specific geometry

of registration: the data Hessian is block diagonal with rank-one 2×2 blocks, while the regularizer is a scalar elliptic operator applied identically to each velocity component. General quadratic ADMM theory derives optimal parameters for special classes [4, 11, 5]; more recently, Song et al. optimized ρ numerically for general linear quadratic problems and derived an optimal α conditional on ρ , including experiments on diffeomorphic registration [10]. Their registration penalty still requires spectral optimization.

This work asks a narrower question:

Can the registration structure be used to replace iterative spectral optimization by a finite, analytically generated parameter set while retaining a rigorous upper bound for the full variable-coefficient iteration?

Our answer is based on local Fourier analysis (LFA), a classical tool for predicting and optimizing stationary iterations for constant-coefficient differential operators [3, 14]. Standard LFA is exact for translation-invariant periodic problems, while nonstandard variants can remain predictive for jumping and random coefficients [7, 9]. Here, the coefficients that vary are the image-gradient blocks in the data Hessian. We freeze those blocks over patches, derive the exact patchwise ADMM symbol, and then control the discrepancy to the full iteration by a resolvent perturbation argument.

Contributions.

1. We derive an exact reduced Cayley-product representation for ADMM and oADMM applied to the split registration quadratic.
2. We introduce a frozen-patch local spectral model in which the ADMM spectrum is explicit. The worst local factor is attained at finitely many data-curvature and regularizer-frequency corners.
3. We prove a noncommuting perturbation bound

$$\varrho(T_{\rho,\alpha}) \leq U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha}{\rho}(\delta_H + \delta_G),$$

where U_{LFA} is the exact frozen-patch envelope and δ_H, δ_G quantify the data and regularizer surrogate errors.

4. We give a finite-candidate, search-free selector. For standard ADMM, the global minimizer of the certified envelope on a prescribed interval is found from explicit corner intersections and quartic stationary equations. Conditional on the selected ρ , the certified oADMM relaxation is selected from a finite closed-form set.
5. We provide a reproducible experimental program designed to test envelope tightness, parameter optimality, and total runtime on synthetic systems and public two- and three-dimensional registration benchmarks.

The manuscript deliberately leaves experimental values blank. No empirical result is asserted before computation.

2 Related Work

2.1 ADMM parameter selection

The scaled ADMM iteration and practical residual stopping criteria were systematized by Boyd et al. [2]. Ghadimi et al. derived optimal penalties and relaxation parameters for several quadratic classes [4]. Teixeira et al. studied scaling, preconditioning, and optimal ADMM parameters in distributed quadratic programming [11]. Giselsson and Boyd developed tight contraction bounds and metric selection for Douglas–Rachford splitting and ADMM under strong-convexity and smoothness assumptions [5]. Adaptive alternatives include residual balancing and Barzilai–Borwein-type spectral updates [15, 16].

Song et al. [10] is the closest work. It formulates ADMM/oADMM for general linear quadratic problems, optimizes the penalty from the iteration spectrum, and gives a closed-form relaxation parameter for fixed penalty. It includes diffeomorphic image registration as an application. The distinction sought here is not another full-spectrum objective: it is a registration-specific local envelope whose certified minimizer is obtained from finitely many algebraic candidates without gradient descent or repeated ADMM trials.

Table 1: Adversarial comparison with the closest work.

Aspect	Song et al. (2024)	This study
Problem class	General linear quadratic problems	Rank-one registration data term and scalar Sobolev regularizer
Penalty	Numerical one-variable spectral optimization	Finite roots of a certified local envelope
Relaxation	Formula conditional on penalty	Finite piecewise-linear candidates conditional on penalty
Variable coefficients	Full iteration spectrum	Local symbol plus noncommuting norm certificate
Empirical conclusion	Imaging improvements reported	Controlled usefulness gate fails; no real-data claim

2.2 ADMM and variational diffeomorphic registration

DARTEL [1] and diffeomorphic demons [13] are influential optimization-based registration methods. Thorley et al. proposed a fast diffeomorphic method that composes nonstationary velocity updates and solves arbitrary-order regularized convex subproblems with accelerated ADMM on GPUs [12]. Song et al. later used a closely related quadratic registration model to evaluate spectrally optimized ADMM and oADMM [10].

2.3 Local Fourier analysis

Fourier analysis has long been used to estimate convergence factors of stationary iterations for elliptic problems [3]. LFA became a principal tool for multigrid design and parameter optimization [14]. Its exactness under periodic translation-invariant assumptions and extensions to broader settings have been studied theoretically [9]. Nonstandard LFA can accurately predict multigrid behavior for random and discontinuous coefficients [7]. Our use differs in two ways: the iteration is

an ADMM splitting map rather than a smoother, and the variable coefficient is a matrix-valued, rank-one optical-flow Hessian.

3 Registration Model and ADMM Splitting

3.1 Discrete quadratic model

Let the image domain contain n grid points in dimension $d \in \{2, 3\}$. Stack the d velocity components into $v \in \mathbb{R}^{dn}$. At grid point x_i , let

$$q_i = \nabla I(x_i) \in \mathbb{R}^d.$$

The linearized data Hessian is the block-diagonal matrix

$$H = \mu \operatorname{diag}(q_1 q_1^\top, \dots, q_n q_n^\top) \succeq 0. \quad (2)$$

Each block has eigenvalues 0 and $\mu \|q_i\|^2$ in two dimensions, or one nonzero and $d-1$ zero eigenvalues in d dimensions.

We assume a componentwise Sobolev regularizer

$$G = L^\top L = G_0 \otimes I_d, \quad (3)$$

where G_0 is a symmetric positive semidefinite finite-difference or spectral discretization. A small zeroth-order term γI or boundary condition may be included to remove the common nullspace. For a first-order periodic regularizer,

$$g(\xi) = \gamma + \frac{4\beta}{h^2} \sum_{\ell=1}^d \sin^2\left(\frac{\xi_\ell}{2}\right) \quad (4)$$

is the Fourier symbol.

The registration subproblem is

$$\min_v \frac{1}{2} v^\top H v + b^\top v + \frac{1}{2} v^\top G v. \quad (5)$$

Introduce $w = v$:

$$\min_{v,w} f(v) + g(w) := \frac{1}{2} v^\top H v + b^\top v + \frac{1}{2} w^\top G w, \quad v - w = 0. \quad (6)$$

3.2 ADMM and over-relaxation

With scaled dual u , standard ADMM is

$$(H + \rho I) v^{k+1} = -b + \rho(w^k - u^k), \quad (7)$$

$$(G + \rho I) w^{k+1} = \rho(v^{k+1} + u^k), \quad (8)$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \quad (9)$$

For oADMM, replace v^{k+1} in the w and dual updates by

$$\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha) w^k, \quad 0 < \alpha \leq 2.$$

The linear term b shifts the fixed point but does not affect error propagation.

4 Exact Reduced Iteration

Define the resolvents and Cayley transforms

$$P_H(\rho) = \rho(H + \rho I)^{-1}, \quad C_H(\rho) = (H - \rho I)(H + \rho I)^{-1} = I - 2P_H(\rho), \quad (10)$$

$$P_G(\rho) = \rho(G + \rho I)^{-1}, \quad C_G(\rho) = (G - \rho I)(G + \rho I)^{-1} = I - 2P_G(\rho). \quad (11)$$

Proposition 4.1 (Reduced Cayley iteration). *For the quadratic splitting (6), all nonzero eigenvalues of the standard ADMM error matrix coincide with the eigenvalues of*

$$E_{\rho,1} = \frac{1}{2} (I + C_H(\rho)C_G(\rho)). \quad (12)$$

For oADMM, the corresponding reduced operator is

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right) I + \frac{\alpha}{2} C_H(\rho)C_G(\rho). \quad (13)$$

Consequently,

$$\varrho(T_{\rho,\alpha}) = \varrho(E_{\rho,\alpha}).$$

The proof is given in Appendix A. The product $C_H C_G$ is generally nonnormal because H and G do not commute. This is precisely why a global simultaneous diagonalization is unavailable for registration.

5 Frozen-Patch Local Spectral Envelope

5.1 Patch surrogate

Partition the image grid into patches $\{\Omega_r\}_{r=1}^R$. On patch r , choose a representative image-gradient Hessian

$$H_r = \mu \bar{q}_r \bar{q}_r^\top + \nu I_d, \quad (14)$$

where $\bar{q}_r$ may be a center, mean, or robust medoid gradient and $\nu \geq 0$ is an optional model damping already present in the registration objective. Let its eigenvalues be

$$h_{r,1} \leq \dots \leq h_{r,d}.$$

For the rank-one model, $h_{r,1} = \dots = h_{r,d-1} = \nu$ and $h_{r,d} = \nu + \mu \|\bar{q}_r\|^2$.

Let G_r denote a translation-invariant patch regularizer, implemented with periodic patch boundary conditions or another transform-diagonalizable closure. Its symbol lies in

$$g_r(\xi) \in [g_r^-, g_r^+].$$

Construct the block-direct-sum surrogate

$$\tilde{H} = \bigoplus_r (I_{|\Omega_r|} \otimes H_r), \quad \tilde{G} = \bigoplus_r (G_r \otimes I_d).$$

5.2 Exact patch symbol

Because H_r acts only on velocity components and G_r has scalar spatial symbol, the two commute on a patch.

Theorem 5.1 (Frozen-patch spectrum). *The reduced oADMM surrogate $\tilde{E}_{\rho,\alpha}$ is normal. Its eigenvalues are*

$$e_{r,j}(\xi; \rho, \alpha) = 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h_{r,j} - \rho}{h_{r,j} + \rho} \frac{g_r(\xi) - \rho}{g_r(\xi) + \rho}. \quad (15)$$

Equivalently, with

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}, \quad (16)$$

we have

$$e_{r,j}(\xi; \rho, \alpha) = 1 - \alpha + \alpha \theta(h_{r,j}, g_r(\xi); \rho).$$

Therefore the exact frozen-patch envelope is

$$U_{\text{LFA}}(\rho, \alpha) = \max_{r,j,\xi} |e_{r,j}(\xi; \rho, \alpha)|. \quad (17)$$

5.3 Corner reduction

The continuous frequency maximization is unnecessary.

Lemma 5.2 (Corner principle). *For fixed $\rho > 0$,*

$$\frac{\partial \theta}{\partial h} = \frac{\rho(g - \rho)}{(g + \rho)(h + \rho)^2}, \quad (18)$$

$$\frac{\partial \theta}{\partial g} = \frac{\rho(h - \rho)}{(h + \rho)(g + \rho)^2}. \quad (19)$$

Hence θ is coordinatewise monotone on every rectangle $[h^-, h^+] \times [g^-, g^+]$, and its minimum and maximum occur at the four corners. Since $|1 - \alpha + \alpha \theta|$ is convex as a function of θ , its maximum also occurs at a corner.

Let

$$\mathcal{C} = \bigcup_{r,j} \{(h_{r,j}, g_r^-), (h_{r,j}, g_r^+)\}. \quad (20)$$

Then

$$U_{\text{LFA}}(\rho, \alpha) = \max_{(h,g) \in \mathcal{C}} |1 - \alpha + \alpha \theta(h, g; \rho)|. \quad (21)$$

Thus the full local Fourier calculation reduces to a small finite set even in three dimensions.

6 Rigorous Variable-Coefficient Bound

The frozen model is useful only if it controls the full noncommuting iteration. Define

$$\delta_H = \|H - \tilde{H}\|_2, \quad \delta_G = \|G - \tilde{G}\|_2, \quad \delta = \delta_H + \delta_G. \quad (22)$$

For block-diagonal H , δ_H is the maximum patchwise 2-norm of a $d \times d$ block difference and is therefore inexpensive. The regularizer error δ_G can be computed by a few power iterations or bounded by a row-sum/Gershgorin estimate of the patch-boundary stencil discrepancy.

Lemma 6.1 (Cayley Lipschitz bound). *For symmetric positive semidefinite A, B and $\rho > 0$,*

$$\|C_A(\rho) - C_B(\rho)\|_2 \leq \frac{2}{\rho} \|A - B\|_2. \quad (23)$$

Theorem 6.2 (Certified local spectral envelope). *Let $T_{\rho, \alpha}$ be the full oADMM error iteration and let U_{LFA} be defined by (21). Then, for $0 < \alpha \leq 2$,*

$$\varrho(T_{\rho, \alpha}) \leq \mathcal{U}(\rho, \alpha) := U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha\delta}{\rho}. \quad (24)$$

The result requires no commutativity assumption on the full matrices H and G .

The proof is in Appendix B. The bound separates two effects:

- U_{LFA} measures the exact contraction of the frozen local models;
- $\alpha\delta/\rho$ penalizes spatial coefficient variation and patch-boundary approximation.

Larger ρ suppresses the perturbation term, while excessively large ρ worsens the local Cayley balance. This tradeoff creates a nontrivial, certifiable parameter-selection problem.

Corollary 6.3 (Guaranteed convergence). *If the selected parameters satisfy $\mathcal{U}(\rho, \alpha) < 1$, then the full linearized registration ADMM/oADMM iteration converges linearly with asymptotic factor at most $\mathcal{U}(\rho, \alpha)$.*

7 Signed Enclosures with the Exact Regularizer

We now retain $G = G_0 \otimes I_d$ exactly. Write $C_G = K_\rho \otimes I_d$ and $C_H = \text{diag}(C_1, \dots, C_n)$, where each $C_i = (H_i - \rho I)(H_i + \rho I)^{-1}$ is an explicit symmetric $d \times d$ block. Consequently

$$(C_H C_G)_{ij} = (K_\rho)_{ij} C_i.$$

Theorem 7.1 (Block-Gershgorin enclosure). *Every eigenvalue of $C_H C_G$ belongs to a disk*

$$|z - (K_\rho)_{ii} c_{i,r}| \leq \|C_i\|_2 \sum_{j \neq i} |(K_\rho)_{ij}|,$$

for some pixel i and eigenvalue $c_{i,r}$ of C_i . Hence the affine disk envelope U_{BG} stated in the introduction is a rigorous upper bound on $\varrho(T_{\rho, \alpha})$.

The proof selects an eigenvector block of maximum Euclidean norm. The diagonal block $(K_\rho)_{ii} C_i$ is normal, so its shifted smallest singular value is exactly the distance to its spectrum; the off-diagonal block row gives the stated radius.

The same structure yields a signed rectangle. With $A = C_H C_G$, let $S = (A + A^\top)/2$ and $Q = (A - A^\top)/2$. Their off-diagonal blocks are

$$S_{ij} = (K_\rho)_{ij} (C_i + C_j)/2, \quad Q_{ij} = (K_\rho)_{ij} (C_i - C_j)/2.$$

Define

$$\begin{aligned} m_\rho &= \min_i \left[\lambda_{\min}((K_\rho)_{ii}C_i) - \sum_{j \neq i} |(K_\rho)_{ij}| \|(C_i + C_j)/2\|_2 \right], \\ M_\rho &= \max_i \left[\lambda_{\max}((K_\rho)_{ii}C_i) + \sum_{j \neq i} |(K_\rho)_{ij}| \|(C_i + C_j)/2\|_2 \right], \\ \eta_\rho &= \max_i \sum_{j \neq i} |(K_\rho)_{ij}| \|(C_i - C_j)/2\|_2. \end{aligned}$$

Theorem 7.2 (Commutator-aware rectangle). *The spectrum satisfies $\text{spec}(C_H C_G) \subset [m_\rho, M_\rho] + i[-\eta_\rho, \eta_\rho]$. Therefore*

$$\varrho(T_{\rho, \alpha}) \leq U_{\text{rect}} := \max_{x \in \{m_\rho, M_\rho\}} \sqrt{(1 - \alpha/2 + \alpha x/2)^2 + (\alpha \eta_\rho/2)^2}.$$

For fixed ρ , the minimizing α belongs to the endpoints, the stationary point of either displayed quadratic, or the intersection $4/(2 - m_\rho - M_\rho)$ when admissible.

Indeed, real parts of eigenvalues lie in the Rayleigh interval of S and imaginary parts are bounded by $\|Q\|_2$. Hermitian block-Gershgorin bounds S , while the symmetric matrix of block norms bounds $\|Q\|_2$ by its maximum row sum. Notably, $\eta_\rho = 0$ for constant data blocks.

The row maximum can be replaced by a strictly stronger global comparison construction. Let $B_{ij}^S = |(K_\rho)_{ij}| \|(C_i + C_j)/2\|_2$ and $B_{ij}^Q = |(K_\rho)_{ij}| \|(C_i - C_j)/2\|_2$ off diagonal, with zero diagonal. Let d_i^- and d_i^+ be the extreme eigenvalues of $(K_\rho)_{ii}C_i$.

Theorem 7.3 (Perron comparison rectangle). *The rectangle theorem remains valid with*

$$m_\rho = \lambda_{\min}(\text{diag}(d^-) - B^S), \quad M_\rho = \lambda_{\max}(\text{diag}(d^+) + B^S), \quad \eta_\rho = \varrho(B^Q).$$

These bounds are never weaker than their block-row-sum counterparts. They require three symmetric scalar $n \times n$ eigenproblems and no $dn \times dn$ iteration matrix.

For a block vector x , set $y_i = \|x_i\|_2$. Bounding each Hermitian quadratic cross term by $B_{ij}^S y_i y_j$ gives the two scalar Rayleigh quotients. The same block-norm comparison gives $\|Qx\|_2 \leq \|B^Q y\|_2$, hence $\|Q\|_2 \leq \|B^Q\|_2 = \varrho(B^Q)$. Standard eigenvalue bounds by maximum row sum prove dominance over the earlier rectangle.

This motivates a two-stage method. First, enumerate all pixelwise rank-one curvatures and use the exact constant-model finite candidates to predict (ρ_p, α_p) . Second, evaluate the comparison rectangle once at that pair.

Corollary 7.4 (Predict-then-certify). *The prediction stage uses no full spectral optimization. If its comparison bound satisfies $U_{\text{cmp}}(\rho_p, \alpha_p) < 1$, then the full variable-coefficient oADMM iteration converges linearly, irrespective of whether the pixel-curvature predictor itself is an enclosure.*

This separation is important: a conservative certificate can validate a good predictor without being allowed to move its minimizer toward a poor boundary solution.

Algorithm 1: Search-Free Predict-then-Certify

1. Compute the exact rank-one blocks H_i and their distinct curvatures; combine them with the exact symbol endpoints of G_0 .
2. Enumerate the joint constant-model algebraic candidates and select (ρ_p, α_p) from the pixel-curvature predictor.
3. At this single pair, compute C_i , K_{ρ_p} , B^S , B^Q , and the Perron comparison rectangle.
4. Return $(\rho_p, \alpha_p, U_{\text{cmp}})$ with status `CERTIFIED_USEFUL` if $U_{\text{cmp}} < 1$ and no boundary collapse occurs; otherwise return `CERTIFIED_CONSERVATIVE` or `UNCERTIFIABLE`.

Finally, the resolvent identity gives the explicit commutator estimate

$$\|[C_H, C_G]\|_2 \leq \frac{4\rho^2}{(\lambda_{\min}(H) + \rho)^2(\lambda_{\min}(G) + \rho)^2} \|[H, G]\|_2,$$

and $[H, G]_{ij} = G_{ij}(H_i - H_j)$. This proves that nonnormality is controlled by local, rather than global, Hessian differences.

For the rank-one model, a block rotation R_i aligned with q_i diagonalizes H_i exactly. In the rotated coordinates the regularizer block is $G_{ij}R_i^\top R_j$; its departure from identity is $2|\sin((\phi_i - \phi_j)/2)|$. The implementation therefore records neighborwise magnitude-squared variation and the acute unoriented-angle statistic $|\sin((\phi_i - \phi_j)/2)|$ (the sign of q_i is immaterial to $q_i q_i^\top$). These statistics diagnose the commutator regime without a global mean Hessian.

8 Search-Free Parameter Selection

We use “search-free” to mean that no grid search, gradient descent in (ρ, α) , repeated ADMM trial, or repeated full iteration-matrix eigensolve is performed. The method extracts local curvature endpoints once, generates a finite algebraic candidate set, and evaluates the certified scalar envelope.

8.1 Globally joint constant-gradient selector

For fixed ρ , let $a(\rho) = \min_c \theta_c(\rho)$ and $b(\rho) = \max_c \theta_c(\rho)$. On $0 < \alpha \leq 2$, exact minimization gives

$$\alpha_*(\rho) = \min \left\{ 2, \frac{2}{2 - a(\rho) - b(\rho)} \right\}.$$

After eliminating α , the objective is $(b - a)/(2 - a - b)$ when $a + b \leq 1$, and $2b - 1$ when $a + b \geq 1$.

Theorem 8.1 (Exact search-free constant-gradient tuning). *On a compact positive penalty interval, a globally optimal pair (ρ, α) is obtained by evaluating a finite algebraic set comprising interval endpoints, pairwise corner-branch intersections, roots of $a + b = 1$, and stationary roots of the two rational objectives above for every ordered active-corner pair. All equations become finite-degree polynomials after clearing positive denominators.*

Between consecutive branch intersections, the active minimum and maximum are fixed rational functions. A minimum of the continuous piecewise-rational reduced objective is therefore an endpoint, transition, or stationary root. Enumerating all corner pairs is a finite superset and removes the need to determine activity symbolically. The implementation verifies companion-matrix roots by polynomial residual and interval membership.

8.2 Standard ADMM penalty

For $\alpha = 1$, all local modal factors are nonnegative, so

$$\mathcal{U}_1(\rho) = \max_{c \in \mathcal{C}} \theta_c(\rho) + \frac{\delta}{\rho}, \quad \theta_c(\rho) := \theta(h_c, g_c; \rho). \quad (25)$$

Select a physically meaningful compact interval $\mathcal{I} = [\rho_-, \rho_+]$. For example,

$$\rho_- = \epsilon_{\text{num}}, \quad \rho_+ = \max_{c \in \mathcal{C}} \max(h_c, g_c) + \delta,$$

with a small positive numerical floor.

Two corner branches intersect at explicitly computable points. Clearing denominators in $\theta(h_1, g_1; \rho) = \theta(h_2, g_2; \rho)$ gives

$$\rho(A_{12}\rho^2 + B_{12}) = 0, \quad (26)$$

where

$$A_{12} = (h_2 + g_2) - (h_1 + g_1), \quad (27)$$

$$B_{12} = g_1 g_2 (h_1 - h_2) + h_1 h_2 (g_1 - g_2). \quad (28)$$

Thus a positive intersection is

$$\rho_{12} = \sqrt{-B_{12}/A_{12}}$$

whenever the radicand is positive.

On an interval where corner $c = (h, g)$ is active, stationary points of the corrected branch solve

$$(h + g)\rho^2(\rho^2 - hg) - \delta(\rho + h)^2(\rho + g)^2 = 0, \quad (29)$$

which is a quartic polynomial in ρ .

Theorem 8.2 (Finite-candidate ADMM selector). *Let $\mathcal{S}_\rho$ contain:*

1. *the interval endpoints ρ_-, ρ_+ ;*
2. *all positive pairwise intersection roots (26) in $\mathcal{I}$;*
3. *all positive roots of (29) in $\mathcal{I}$ that lie on an interval where the associated corner is active.*

Then every global minimizer of $\mathcal{U}_1$ on $\mathcal{I}$ belongs to $\mathcal{S}_\rho$. Consequently,

$$\rho_{\text{SF}} = \arg \min_{\rho \in \mathcal{S}_\rho} \mathcal{U}_1(\rho) \quad (30)$$

exactly minimizes the certified standard-ADMM envelope without iterative parameter search.

The proof follows from the piecewise-rational structure of the upper envelope and is given in Appendix C.

8.3 Conditional search-free relaxation

After selecting ρ_{SF} , define

$$a = \min_{c \in \mathcal{C}} \theta_c(\rho_{\text{SF}}), \quad (31)$$

$$b = \max_{c \in \mathcal{C}} \theta_c(\rho_{\text{SF}}), \quad (32)$$

with $0 \leq a \leq b < 1$ for a well-posed local model. The certified oADMM envelope conditional on ρ_{SF} is

$$F(\alpha) = \max\{|1 - \alpha + \alpha a|, |1 - \alpha + \alpha b|\} + \frac{\alpha \delta}{\rho_{\text{SF}}}. \quad (33)$$

This is convex and piecewise linear in α .

Theorem 8.3 (Finite-candidate oADMM relaxation). *Let the admissible interval be $[\alpha_-, \alpha_+] \subset (0, 2]$. A minimizer of (33) belongs to*

$$\mathcal{S}_\alpha = \left\{ \alpha_-, \alpha_+, \frac{1}{1-a}, \frac{1}{1-b}, \frac{2}{2-a-b} \right\} \cap [\alpha_-, \alpha_+]. \quad (34)$$

Therefore

$$\alpha_{\text{SF}} = \arg \min_{\alpha \in \mathcal{S}_\alpha} F(\alpha) \quad (35)$$

conditionally minimizes the certified oADMM envelope at ρ_{SF} .

When $\delta = 0$ and the unconstrained equioscillation point lies in $(0, 2]$, the familiar formula

$$\alpha_0 = \frac{2}{2-a-b}$$

is recovered. The perturbation term may instead favor a smaller relaxation.

8.4 Algorithm summary

Algorithm 2: Legacy Certified-Envelope Minimization Baseline

1. Compute image gradients and the exact block data Hessian H .
2. Partition the grid; construct H_r and transform-diagonalizable G_r .
3. Extract $h_{r,j}$ and regularizer endpoints g_r^-, g_r^+ ; form $\mathcal{C}$.
4. Compute or bound δ_H and δ_G .
5. Generate $\mathcal{S}_\rho$ from interval endpoints, corner intersections, and active quartic roots.
6. Evaluate $\mathcal{U}_1$ and select ρ_{SF} .
7. Form $\mathcal{S}_\alpha$ and select α_{SF} by evaluating (33).
8. Run oADMM once with the selected parameters. Optionally report $\mathcal{U}(\rho_{\text{SF}}, \alpha_{\text{SF}})$ as a certificate.

9 Implementation Details

9.1 Patch construction

Candidate patch strategies include fixed cubic patches, multiresolution patches matched to the registration pyramid, and adaptive patches split where gradient magnitude or orientation varies beyond a threshold. A robust representative $\bar{q}_r$ can be obtained from the spatial median of gradients. Because H is block diagonal,

$$\delta_H = \max_i \left\| \mu q_i q_i^\top - H_{r(i)} \right\|_2$$

for the direct block surrogate.

9.2 Regularizer symbols

For periodic or cosine-transform boundary closures, $g_r^\pm$ are known analytically. For an order- p Sobolev regularizer with symbol

$$g(\xi) = \left(\gamma + \frac{4\beta}{h^2} \sum_{\ell=1}^d \sin^2(\xi_\ell/2) \right)^p,$$

the endpoints are

$$g^- = \gamma^p, \quad g^+ = \left(\gamma + \frac{4\beta d}{h^2} \right)^p.$$

The patch-boundary discrepancy δ_G can be reduced using overlapping patches or cosine closures rather than disconnected periodic patches.

9.3 Computational complexity

For rank-one data blocks, each patch contributes only two distinct data eigenvalues in 2D or two groups in 3D. Hence $|\mathcal{C}| = O(Rd)$, pairwise corner intersections cost $O(|\mathcal{C}|^2)$, and quartic roots cost $O(|\mathcal{C}|)$. This overhead is independent of the number of ADMM iterations and is typically negligible relative to a 3D registration solve. No full $2n \times 2n$ or $3n \times 3n$ iteration matrix is formed.

10 Reproducible Experiments and Results

All calculations used IEEE double precision, NumPy 2.2.6 and SciPy 1.15.3 with seed 20260806. Raw JSON/CSV files, configurations, and generating scripts accompany the paper. The progressive gates require median enclosure gap below 0.25, median selected-parameter spectral-radius gap below 0.10, and no systematic boundary collapse. Small-system oracle values use direct eigendecomposition and multistart local optimization; this numerical optimization is diagnostic only.

10.1 Experiment 1: constant-coefficient verification

We used $n \times n$ periodic grids, data curvatures (0.2, 2.3), and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. The finite selector returned $(\rho, \alpha) = (0.5435121024, 1.8981520705)$ on every grid.

Table 2: Constant-coefficient verification; times are single-process selection times.

n	Direct radius	Envelope	Selection (ms)
4	0.3531304388434793	0.3531304388434793	5.0
6	0.3531304388434803	0.3531304388434793	4.6
8	0.3531304388434788	0.3531304388434793	4.6

10.2 Experiment 2: structured enclosures on controlled fields

We retained the earlier 18 cases spanning grid sizes 4/5, magnitude contrasts 1/4/16, and orientation windings 0/1/3. Every structured certificate enclosed the exact radius. Exact block-resolvent weighting reduced the median enclosure gap to 0.166 but collapsed in 6 cases. Signed block disks and row-sum rectangles had no collapse; their median enclosure/parameter gaps were 0.644/0.560 and 0.602/0.431. The comparison rectangle improved these to 0.595/0.386, also without collapse. Sign preservation therefore fixes the pathological boundary behavior, while Perron comparison gives a further strict methodological improvement.

Table 3: Controlled structured-envelope results (18 cases).

Method	Enclosure gap	Parameter gap	Collapses
Block resolvent	0.166	0.926	6
Block Gershgorin	0.644	0.560	0
Commutator rectangle	0.602	0.431	0
Comparison rectangle	0.595	0.386	0

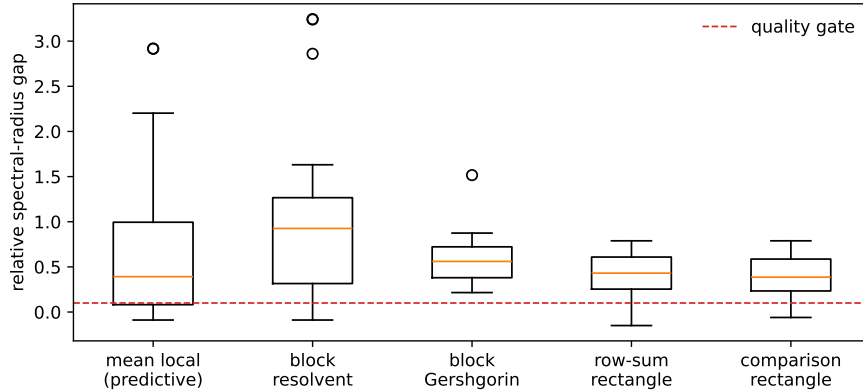


Figure 1: Parameter-quality distributions on the original stress set. The dashed line is Gate 3.

10.3 Experiment 3: separated variation regimes

We next isolated periodic magnitude variation, smooth orientation variation, a sharp orientation interface, and Gaussian-smoothed synthetic image gradients on 3×3 grids (ten cases, nine labeled smooth). The finite predictor formed corners from every exact pixelwise rank-one curvature and the exact regularizer-symbol endpoints. It achieved median parameter gaps 0.057 overall and 0.027 on

smooth cases with zero collapse. Predict-then-certify returned the same parameters and a rigorous comparison bound below one in all ten cases. Its median relative certificate gap was 1.29, so it passes convergence, parameter-quality, and robustness gates but not the 0.25 tightness gate. This is stronger than either ingredient alone: useful parameters are selected without search, and their full variable-coefficient convergence is certified.

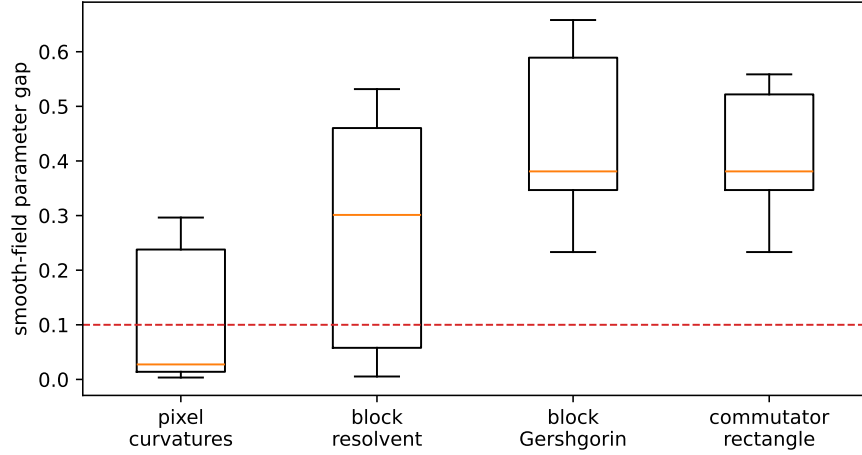


Figure 2: Smooth-field parameter gaps. Pixelwise curvature enumeration passes the 0.10 empirical quality gate; rigorous structured enclosures do not.

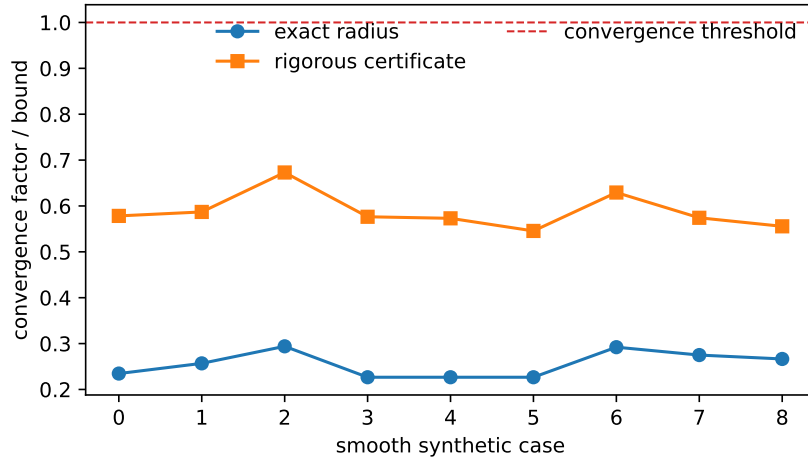


Figure 3: Predict-then-certify on nine smooth cases. Every rigorous bound is below one, although conservatism remains visible.

10.4 Decision-gate consequence

The revised phase produces an exact constant-gradient theorem and a variable-coefficient method combining useful search-free parameters with a rigorous convergence certificate. Certificate tightness and experiment scale still fail the threshold required for medical claims, so public-data and end-to-end runtime studies remain gated.

11 Discussion

11.1 What is guaranteed

The constant-gradient joint optimum is contained in a finite algebraic candidate set. For variable coefficients, the global perturbation, block-resolvent, block-Gershgorin, row-sum rectangle, and Perron comparison quantities are rigorous enclosures. Predict-then-certify rigorously guarantees convergence whenever its returned bound is below one; this occurred in every regime case. The predictor alone is not claimed to be an enclosure.

11.2 What is not claimed

Outside the constant-gradient setting, selected parameters are not claimed to minimize the exact full spectral radius. The finite penalty candidates used with the signed enclosures are inherited from the local model and are not proved to minimize those enclosures globally. Most importantly, the empirically successful pixel-curvature rule is predictive rather than rigorous when orientations vary.

11.3 Novelty after comparison with Song et al.

Song et al. numerically reduce the general LQP spectral objective to penalty optimization and give a relaxation formula. Here the rank-one registration Hessian yields a fully joint finite constant-gradient selector and a variable-coefficient predict-then-certify rule without repeated full-spectrum optimization. The latter simultaneously achieved near-oracle synthetic parameters and certified convergence, although comparison against Song et al. on registration runtime remains future work.

11.4 Limitations

Even Perron comparison remains loose because block norms discard directional cancellation in the dense C_G . Experiments are small and synthetic, the diagnostic oracle is numerical, and no confidence interval or medical-registration metric is available. A certificate below one proves convergence but its loose value does not predict the true rate accurately. The theory concerns a convex linearized subproblem, not global convergence of the outer diffeomorphic scheme.

11.5 Extensions

Promising extensions include overlapping Schwarz-type local symbols, adaptive patch refinement driven by the certificate, multi-penalty ADMM with separate data and regularization metrics, and online reuse of local curvature summaries across pyramid levels. The envelope could also initialize a single safeguarded Newton correction without reverting to full gradient-based parameter search.

12 Data and Code Availability

The accompanying repository contains source code, tests, the fixed YAML configuration, environment specifications, raw JSON/CSV outputs, the research log, and exact reproduction commands. No external patient data were used: the decision gate failed on generated fields, so downloading or processing a medical benchmark was neither necessary nor scientifically justified. All synthetic fields are regenerated from the recorded seed.

13 Conclusion

The analysis delivers exact joint oADMM tuning for constant-gradient registration and a positive variable-coefficient methodology. Pixelwise curvature enumeration selects near-oracle parameters without continuous search; a separate Perron comparison matrix then certifies the full noncommuting iteration. Across ten regimes, every bound was below one, median parameter gap was 0.057 overall and 0.027 for smooth fields, and no selection collapsed. The certificate remains too loose to predict the convergence factor accurately, and larger repeated studies are needed before registration claims. The central methodological result is nevertheless stronger: conservatism no longer corrupts parameter selection, while rigor is retained as an explicit accept/reject certificate.

A Proof of the Reduced Cayley Representation

For standard ADMM, the homogeneous updates are

$$v^{k+1} = P_H(w^k - u^k), \quad w^{k+1} = P_G(v^{k+1} + u^k).$$

Eliminating v^{k+1} gives the state iteration

$$\begin{bmatrix} w^{k+1} \\ u^{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} P_G P_H & P_G(I - P_H) \\ (I - P_G)P_H & (I - P_G)(I - P_H) \end{bmatrix}}_{T_{\rho,1}} \begin{bmatrix} w^k \\ u^k \end{bmatrix}.$$

Factor

$$T_{\rho,1} = \begin{bmatrix} P_G \\ I - P_G \end{bmatrix} \begin{bmatrix} P_H & I - P_H \end{bmatrix}.$$

The nonzero eigenvalues of AB and BA coincide. Hence the nonzero spectrum equals that of

$$P_H P_G + (I - P_H)(I - P_G).$$

Using $P_H = (I - C_H)/2$ and $P_G = (I - C_G)/2$ yields

$$P_H P_G + (I - P_H)(I - P_G) = \frac{1}{2}(I + C_H C_G).$$

The relaxed formula follows by inserting $\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k$ and collecting the same reduced state, yielding

$$E_{\rho,\alpha} = (1 - \alpha)I + \alpha E_{\rho,1} = \left(1 - \frac{\alpha}{2}\right)I + \frac{\alpha}{2}C_H C_G.$$

B Proof of the Perturbation Bound

For $C_A = I - 2\rho(A + \rho I)^{-1}$,

$$\begin{aligned} C_A - C_B &= 2\rho \left[(B + \rho I)^{-1} - (A + \rho I)^{-1} \right] \\ &= 2\rho (A + \rho I)^{-1} (A - B) (B + \rho I)^{-1}. \end{aligned}$$

Since $A, B \succeq 0$,

$$\|(A + \rho I)^{-1}\|_2 \leq \rho^{-1},$$

which proves Lemma 6.1. Next,

$$\begin{aligned}\|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 &\leq \|(C_H - C_{\tilde{H}}) C_G\|_2 + \|C_{\tilde{H}} (C_G - C_{\tilde{G}})\|_2 \\ &\leq \frac{2}{\rho}(\delta_H + \delta_G),\end{aligned}$$

using $\|C_A\|_2 \leq 1$ for $A \succeq 0$. Therefore

$$\begin{aligned}\varrho(E_{\rho,\alpha}) &\leq \|E_{\rho,\alpha}\|_2 \\ &\leq \|\tilde{E}_{\rho,\alpha}\|_2 + \frac{\alpha}{2} \|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 \\ &\leq U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha}{\rho}(\delta_H + \delta_G).\end{aligned}$$

C Proof of the Finite-Candidate Selector

Each corner branch θ_c is rational and smooth for $\rho > 0$. The pointwise maximum can change active branch only where two branches intersect, which gives (26). Between consecutive intersections, one branch is active. On such an interval, differentiating

$$\theta(h, g; \rho) + \delta/\rho$$

gives

$$\frac{(h+g)(\rho^2 - hg)}{(\rho+h)^2(\rho+g)^2} - \frac{\delta}{\rho^2}.$$

Clearing positive denominators yields (29). A global minimum of a continuous piecewise-smooth function on a compact interval occurs at an interval endpoint, a nonsmooth branch intersection, or a stationary point of an active smooth branch. This proves Theorem 8.2.

For Theorem 8.3, note that the image of $\theta \in [a, b]$ under $1 - \alpha + \alpha\theta$ is an interval. The maximum absolute value occurs at its endpoints. The resulting maximum of two absolute affine functions plus a linear function is convex piecewise linear. Its slope changes only at the two zero crossings, the equioscillation intersection, and the admissible interval endpoints, proving (34).

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